

Differential Equation- Applications

1. The volume V of a large spherical snowball decreases as it melts at a rate proportional to its surface area. Find the equation of $V(t)$.
2. Heat is supplied to a body at a constant rate g , and at the same time the body loses heat to the surrounding at a rate proportional to the temperature T exceeds over the room temperature T_R . Find the equation of $T(t)$.
3. A tank initially contains 1000L of pure water. Salt water begins pouring into the tank from a pipe and a stirring blade ensures that it is completely mixed with the pure water. A second pipe draws the water and salt water mixture off at the same rate, so that there is always a total of 1000L in the tank. Let Q grams of salt in the tank at time t and the salt water flowing in the tank at the constant rate of w L/min. Given that the concentration of the salt water is 2 grams of salt per Litre.
 - a. Show that $\frac{dQ}{dt} = -\frac{w}{1000}(Q - 2000)$
 - b. If there is 1 kg of salt in the tank after 5.75 hrs, find w
4. In the manufacturing process of soft drink, a tank initially contains 20 litres of a mixture with consists of grape juice into which 5 kg of sugar has been dissolved. At any given time the sugar is fermented into soft drink at a rate of $4c$ kg/day, where c kg/liter is the concentration of unfermented sugar in the mixture. Assuming that there is no change in the volume of the mixture.
 - a. Show that, if x kg of sugar remains after t days, then $\frac{dx}{dt} = -\frac{x}{5}$.
 - b. Hence find the time taken for the quantity of sugar in the mixture to fall to 0.005 kg. Correct the answer to one decimal place.
 - c. A second tank also initially contains 5 kg of sugar in 20 litres of mixture but, as it is not sealed correctly, the liquid evaporates at a constant rate of 100 ml/day.
 - i. Show that, if y kg of sugar remains after t days, then $\frac{dy}{dt} = -\frac{40y}{200-t}$
 - ii. Hence find the time taken for the sugar content to fall to 0.005 kg in the second tank. Correct the answer to one decimal place.
5. A particle moves in a straight line with initial speed u at time $t = 0$, and with acceleration inversely proportional to $(t + t_0)^3$, where t_0 is positive and constant. Given that $v(0) = u$.
 - a. Show that the speed of the particle approaches to a limiting value.
 - b. If this limiting value is $2u$, show that after time t the particle will have travelled a distance $\frac{ut(2t+t_0)}{t+t_0}$
6. Find the General Solution of the following Differential Equation:
 - a. $x \ln x \frac{dy}{dx} = 1$
 - b. $(xy^2 - y^2 - 4x + 4) \frac{dy}{dx} = x + 3$
 - c. $2 \tan y \frac{dy}{dx} = \sin(x + y) + \sin(x - y)$
 - d. $(x + y) \frac{dy}{dx} = x - y$
 - e. $\frac{dy}{dx} = \sin^2(x - y)$
 - f. $2x^2 \frac{dy}{dx} - x^2 y^2 + 2xy - 1 = 0$
 - g. $3x \frac{dy}{dx} = 3y - x^2 y^2$